EL NIÑO ACTION BRIEFING · 26 OCTOBER 2026

Canyon, Redwood City

Here an El Niño winter means trees on cars and roads, outages that last days, and slides on saturated slopes, worst on the third or fourth storm. The first move is to register and call the older households on the hillside before 15 December.

IMPACT REGION	EXCESS HEAT DEATHS	WINDOW	DOMINANT HAZARD
San Mateo County, CA (inferred)	None published for the US, California, or the county	n/a here; global 6-month figure 239,000 ± 11,000	Trees on cars and roads, multi-day PG&E outages, slides on saturated slopes late in a wet winter

14/14 recommendations checked. 0 confirmed outright, 8 corrected, 0 unverified, 6 contraindicated (the Do-not cards). The five resident steps were checked separately (1 confirmed, 4 corrected). The funding target fell 46% in verification, from $81,480 to $43,980. The Rebuilding Together drainage and tree placement moved to the ceiling, because no evidence was found that it does that work. Exposure: Climate Impact Lab, *451,000 at Risk*, September 2026, pre-peer-review. It publishes no figure for the United States, California, or San Mateo County, and heat is not this place's winter hazard. No population-scaled figure is given here.

EXECUTIVE SUMMARY — AT A GLANCE



37.47°N 122.26°W, United States of America · Natural Earth

- **The risk.** Storms hit here in every ENSO phase. El Niño's mark is the long wet winter: in 1998 San Mateo County had the Bay Area's costliest slides.
- **For you.** Register for SMC Alert at your home address today, with the landline option. It is free, and you get most alerts only if you sign up.
- **The money.** $15,000 floor, $43,980 target, $110,460 ceiling. The cheapest wins are PG&E battery rebates claimed before the last ones run out.
- **Do not.** Never run a generator indoors or in a garage. Two men died at Sea Ranch in January 2023 with one in the hallway.*

For you, as a resident	5 steps	p. 2
What to do	**8** actions	p. 4
Who does what	RACI chart	p. 11
Where money does the most	funding	p. 11
Do not do this	**6** items	p. 13
What has happened here	the record	p. 15
How this was produced	methodology	p. 17
References	sources	p. 18

* Suggested formulation: check with a health worker before sharing.

READ THIS FIRST

EL NIÑO AND THIS PLACE

El Niño does not decide whether storms hit here. It decides how long the wet season lasts. San Mateo County has had 15 storm disaster declarations since 1981, and only four came in El Niño winters (1983, two in 1995, and 1998). The deadliest storm, in January 1982, came in a neutral winter, and the 2023 storms came in a La Niña. El Niño's local signature is the long, saturating winter, as in 1982–83 and 1997–98. Risk peaks on the third or fourth storm, and slides keep moving for days after the rain stops. So every action below is triggered by the storm and the soil, not by the El Niño label.

The request gave a street address inside the City of Redwood City. The briefing is widened to Redwood City's Canyon neighbourhood and the Farm Hill hillside. It includes unincorporated County pockets inside Canyon (Hillcrest, Oak Knoll, and Upland, per a local blog, not a County map), with unincorporated Emerald Lake Hills to the west. Tract 6098 (ACS 2020–24) has 4,491 people: 937 are 65 or over (21%), 407 are 75 or over, and 73 people aged 65 or over live alone. Of 1,670 households, 623 include someone 65 or over. 16% rent, 2.4% are in poverty, 57.8% of homes were built before 1960, and 4.1% speak English less than very well. ZIP 94062 is far larger (75.8 square miles, with Woodside and Emerald Hills), so ZIP figures are labelled as such.

FOR YOU, AS A RESIDENT

5 STEPS

For a resident of this hillside, an El Niño winter means a falling tree on the car or the one road out, days without power, and a slope that gives way late in the season. Most households here can buy their own protection. No role was given with the request, so this section is written for a resident. The ranked actions below are written for the city, the county, and their partners. These are the ones you can take yourself.

1 · Register for SMC Alert at your home address, and add the landline voice option

CONFIRMED FROM ACTION 1

BY WHEN	COST TO YOU	PROTECTS
Now	None	Everyone in the house

SMC Alert is opt-in: the County says that without registering you "can't be reached for the majority of non-life threatening emergencies." Your address lets a slide or road warning for your zone reach you, and the landline still rings when the mobile network is down.

2 · Check your address on PG&E's fire-threat map this week, and claim the battery rebate if you qualify

* Suggested formulation: check with a health worker before sharing.

PARTIALLY CONFIRMED · FROM ACTION 2

BY WHEN	COST TO YOU	PROTECTS
This week; rebate deadline 31 Dec 2026	Roughly $700–1,700 for a battery station after the $300 rebate	Medical devices, phones, lights, and the fridge

If your address is in the High Fire-Threat District, buy a qualifying battery and apply for the $300 rebate right after delivery; about 125 rebates were left across all of PG&E on 15 October, so assume the pool may be gone. If anyone at home uses powered medical equipment, also enrol in Medical Baseline, which gives lower-priced energy, but its doorbell checks come only before fire-season shutoffs, not storm outages.*

3 · Pipe your downspouts to the street, never onto the slope, and learn the signs of a slide

PARTIALLY CONFIRMED · RESIDENT-SPECIFIC

BY WHEN	COST TO YOU	PROTECTS
1 Dec 2026	About $100–300 for pipe work; several hundred to over $1,000 for a geologist (estimates)	The house and the slope under it

Run downspouts through solid pipe to the paved street or a storm drain, as the NRCS hillside guidance advises, because water poured onto a slope is what loosens it. Watch for new cracks, seeps, tilting trees, and doors that stick, and if you see them, call a geologist.

4 · In a High Wind Warning, stay off the roads

PARTIALLY CONFIRMED · FROM ACTION 4

BY WHEN	COST TO YOU	PROTECTS
Every warning, all winter	None	Everyone in the car

Of 407 US deaths from falling trees in 1995–2007, 44% were people in vehicles, and in March 2023 a falling tree killed a van driver on Alpine Road in Ladera, about 9 km away. Sleeping away from rooms under big trees is a sensible precaution, but a weaker one: only 9% of those deaths were in frame houses.

* Suggested formulation: check with a health worker before sharing.

5 · Test or replace your CO alarm, and run any generator only outdoors

PARTIALLY CONFIRMED · FROM DO-NOT 1

BY WHEN	COST TO YOU	PROTECTS
Before the first storm	About $25–50 per alarm (estimate)	Everyone asleep in the house

California already requires CO alarms in many homes; check yours has a battery, because a mains-only alarm dies in the outage. Keep any generator outdoors, at least 20 feet from any occupied building, which is the Sonoma Sheriff's advice after two men died at Sea Ranch in 2023.*

Optional, only if your lot sits on Cordilleras Creek or another channel. An NFIP flood policy bought by about 10 November covers you by December, because new policies start after a 30-day wait. FEMA's authority to write new policies expires on 11 December 2026 unless Congress extends it. It covers flood, and mudflow that crosses two or more acres or two or more properties. It never covers a landslide ("saturated soil mass moving by liquidity down a slope" is excluded), and standard homeowners policies also exclude earth movement (California Department of Insurance, February 2025).

What to ask of the people in charge

- Ask **San Mateo County DEM** who answers for your street, the City or the County, and which shelter is named for it (actions 5 and 6).
- Report a leaning tree over your road to **Redwood City Public Works**, or to **PG&E** if it is on a line (action 3).

WHAT THIS PLACE SHOULD DO — RANKED

Ranked by expected lives saved per unit of effort before the peak. Backing lenses are shown per action; an action several lenses reached independently is the stronger bet. No storm death is on record on Canyon Road or Farm Hill itself. The ranking follows who would die first if one

* Suggested formulation: check with a health worker before sharing.

happened: older people alone in an outage, and people driving under trees.

1. Register and call the older households before the first storm

PARTIALLY CONFIRMED | PUBLIC HEALTH · FRONTLINE · EMERGENCY MGR · PHILANTHROPIST · SKEPTIC | INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Redwood City CERT for city addresses, run with County OES. Unincorporated pockets: fire agency unresolved (see action 6)	15 Dec 2026	Volunteer time; $10,000 for CERT and WPV-Ready rosters (estimate, see funding)	Deaths of isolated older and device-dependent people in multi-day outages

Door-knock the roughly 620 households with someone 65 or over, including the 73 older people who live alone. Register each for SMC Alert at the home address, with the landline voice option. SMC Alert is opt-in; Wireless Emergency Alerts still reach phones, but only for life-threatening events. Ask "what in this house needs power?" Give each household a named check-in contact and a "car buddy" on another access road. For unincorporated pockets and Emerald Hills, the fire agency could be County Fire/CAL FIRE, Redwood City Fire under contract, or Woodside Fire's WPV-Ready (coordinator Selena Brown, 650-851-1594). Action 6 settles which. Medical Baseline's extra warnings and doorbell check apply only to fire-weather shutoffs (PSPS), not winter storm outages; its value is a lower-priced energy allowance.

* Suggested formulation: check with a health worker before sharing.

Evidence: Lurie et al. (*AJKD* 2015) followed about 13,000 Medicare dialysis patients through Hurricane Sandy. Those dialysed early had 28% lower odds of death within 30 days (OR 0.72). It is observational, and it transfers weakly: ZIP 94062 has 11 dialysis patients (emPOWER, Medicare only, across 75.8 square miles). The same file counts 169 people on electricity-dependent equipment and 14 on oxygen.

2. Claim PG&E's battery rebate before it runs out

PARTIALLY CONFIRMED | ECONOMIST · PHILANTHROPIST · PUBLIC HEALTH | INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Redwood City CERT runs the drive; PG&E pays the rebate; CID San Mateo runs any pool	**This week; rebate deadline 31 Dec 2026**	**Rebate paid by PG&E; $5,000 drive and a $28,980 pool (estimates, see funding)**	**Loss of oxygen, medical devices, light, and heat in multi-day outages**

A point query of the CPUC layer puts one address in the Canyon neighbourhood in High Fire-Threat District Tier 2, so any PG&E account in that tier qualifies, with no medical requirement. The edge of Tier 2 runs about 150–200 m east, so check each address on PG&E's map. The rebate is up to $300, plus up to $200 for CARE/FERA customers (rare here). One per account, qualified products only, apply after delivery, by 31 Dec 2026 or 12 months after purchase. PG&E said "approximately 125 rebates remain as of October 15, 2026" across its whole territory, so act this week or assume it is gone. PG&E's Portable Battery Program is closed for 2026. CERT helps device households file. Its $5,000 drive is released only after confirming rebates remain.

* Suggested formulation: check with a health worker before sharing.

For device households who miss it: a battery pool through CID San Mateo (Center for Independence of Individuals with Disabilities), PG&E's disability-access partner for the county. Its partner batteries are issued only for PSPS shutoffs, so storm outages need a separate pool: 20 stations × $1,199 + $5,000 = $28,980. This is conditional on CID agreeing to run one county-wide pool with Belmont's (see funding).

3. Survey hazard trees and clear drains before the soil saturates

PARTIALLY CONFIRMED | EMERGENCY MGR · HISTORIAN · FRONTLINE · PHILANTHROPIST | INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Redwood City Public Works (city streets); County Public Works (unincorporated roads)	15 Dec 2026	Existing crews; arborist contract $15,000–30,000 (estimate)	A tree onto a car, a house, or the only road out

Survey roadside trees on Canyon, Edgewood, Cordilleras, Farm Hill, and the Emerald Lake Hills roads. Check roots and saturation, not only the canopy. The oak that fell into a Ferndale Way house in Emerald Hills on 19 December 2019 had recently been trimmed, and it fell on saturated ground. Report trees on lines to PG&E. Clear culverts by 15 December.

Precedent: on 21 March 2023 a tree killed a 29-year-old van driver on Alpine Road near La Cuesta Road in Ladera, about 9 km away. No study measures how many deaths roadside tree removal prevents; the case rests on the mechanism and on Schmidlin (action 4). The crews belong to two agencies, so the split depends on action 6.

4. One household rule for wind warnings: stay off the roads

PARTIALLY CONFIRMED | SKEPTIC · PUBLIC HEALTH · EMERGENCY MGR | INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
San Mateo County DEM, pushed with each NWS warning	Message ready by 30 Nov; on each High Wind Warning	None	Deaths of drivers and passengers under falling trees

* Suggested formulation: check with a health worker before sharing.

When the National Weather Service issues a High Wind Warning, stay off the roads. Schmidlin (*Natural Hazards* 2009, 50:13–25) counted 407 US deaths from wind-thrown trees in 1995–2007. 44% were in vehicles and 9% in frame houses. The County should push the rule through SMC Alert with each warning.

Transferability: the data are national, but they match the local case (Ladera, 2023). Sleeping away from rooms under big trees is a weaker, secondary precaution, since houses account for few of the deaths.

5. Slide warnings that people act on

PARTIALLY CONFIRMED | EMERGENCY MGR · SKEPTIC · PUBLIC HEALTH | INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
San Mateo County DEM, with the Sheriff	**Zones, wording, and named shelter by 1 Dec; then on each watch**	**Staff time**	**People staying in a slide path because the warning was vague or there was nowhere to go**

No NWS or USGS debris-flow or landslide watch exists for unburned Bay Area slopes; debris-flow warnings use burn-scar thresholds. So the trigger is an NWS Flood or Flash Flood Watch on saturated soils, or high season-to-date rain. Issue Genasys zone messages (more than 300 Peninsula zones; Redwood City codes run RWC-E###; the zone for this neighbourhood was not found). Use the state's Evacuation Warning and Evacuation Order terms, with a plain consequence. Name the shelter before any order. On Woodside's Patrol Road on 22 March 2023, about 30 homes got a "highly recommended evacuation" and "An evacuation location has not been set up yet." Keep advisories up for days after rain; the Patrol Road advisory ended after five dry days. Draw zones on slide and channel hazard. Teach the physical signs: new cracks, seeps, tilting trees, and doors that stick.

* Suggested formulation: check with a health worker before sharing.

Montecito 2018: 19 of the 23 dead lived in the voluntary zone (secondary source). Goto et al. (*IJDRR* 2021) found that being in the voluntary zone was the main reason people stayed, and the zones had been drawn from the Thomas Fire, not from debris-flow runout. Montecito was post-fire, so it compares here on communication only.

6. Settle who answers on each street

PARTIALLY CONFIRMED | EMERGENCY MGR · FRONTLINE | INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
San Mateo County DEM, with Redwood City Fire, the Sheriff, and the County fire authority	30 Nov 2026	Staff time	Delay at the City–County seam when a road closes or a house slides

Walk the neighbourhood with the County fire-jurisdiction map. Set who closes roads, who evacuates, and who red-tags on each street. Publish a jurisdiction card at the churches and the Canyon Inn. Bonfare Market is unverified as open, so leave it off until someone confirms it.

Why it matters: unincorporated County pockets sit inside a city neighbourhood, and the fire agency for those pockets and Emerald Hills is unresolved. A County source names Redwood City Fire under contract for Emerald Hills; the lens briefs named Woodside Fire. Action 1 needs this answer to name its owners.

7. Clear the known choke points: Cordilleras Creek and Edgewood Road

PARTIALLY CONFIRMED | HISTORIAN · FRONTLINE | INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Redwood City Public Works, with OneShoreline and County Public Works	Creek this autumn; Edgewood slope inspected by 1 Dec 2026	Not costed	A repeat of the 2022 creek flood and the 2023 road closure

* Suggested formulation: check with a health worker before sharing.

On 31 December 2022, log dams of debris from I-280 flooded at least four homes in Centennial; the creek ran about 10 feet deep in the channel. Owners must clear the creek under Ordinance 2090. The City said it "wants to" do annual maintenance. That is an aim, not a promise: in February 2024 it was still "in question," and no access agreement was found. Ask for one agency work order, for example under OneShoreline's 2024 maintenance filing, which covers stretches of Cordilleras Creek. On Edgewood Road (I-280 to Crestview), debris was removed in January 2023 and the road reopened in December 2023. The County status reads "No additional repair planned." A debris fence and weep holes were only listed as possible work, so inspect the slope and any drainage by 1 December.

Corrections: the 2023 Edgewood work is not a documented fix, only debris removal and a reopening. A second Edgewood closure on 22 March 2023 is unverified.

8. Reach homebound people with food when the road or the power is out

PARTIALLY CONFIRMED | FOOD SUPPLY · PUBLIC HEALTH | INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Peninsula Volunteers (Meals on Wheels); San Mateo County HSA	**1 Dec 2026**	**From Belmont's $40,000 Peninsula Volunteers placement, or costed separately**	**Homebound clients going without food through a multi-day outage**

Peninsula Volunteers stages Meals on Wheels storm packs for hillside clients by 1 December. Draw them from the Belmont briefing's $40,000 Peninsula Volunteers placement (300 shelf-stable kits), or cost them separately; do not fund them twice. San Mateo County HSA prints the CalFresh replacement notice: report food loss within 10 days (form CF 303; 1-800-223-8383). HSA should also give CDSS the affected ZIPs fast. The state, not the county, requests mass replacement from USDA, and San Mateo was in the January 2023 automatic replacement.

The food signal here is very weak. Median household income in the tract is at the census cap ($250,001), so this action serves a small number of homebound clients, not the neighbourhood.

* Suggested formulation: check with a health worker before sharing.

WHO DOES WHAT

RACI CHART

R = Responsible (does the work) · **A** = Accountable (one per action; signs off and answers for it) · **C** = Consulted (asked before) · **I** = Informed (told after). Built from the owners named in the verified actions. No one here has been contacted or has agreed to a role; every row is a proposal.

Action	County DEM	Sheriff	RWC CERT	RWC Fire	County fire authority	RWC Public Works	County Public Works	PG&E	CID San Mateo	OneShoreline	Peninsula Volunteers	County HSA	Households
1. Register and call older households	**R**		**A/R**	**C**	**R**			**I**					**R**
2. PG&E battery rebate drive and pool	**I**		**A/R**					**R**	**R**				**R**
3. Hazard trees and drains	**I**					**A/R**	**R**	**R**					**I**
4. Stay off roads in a wind warning	**A/R**	**C**	**I**										**R**
5. Slide warnings and a named shelter	**A/R**	**R**		**C**	**C**								**I**
6. Who answers on each street	**A/R**	**R**		**R**	**R**	**C**	**C**						
7. Cordilleras Creek and Edgewood Rd						**A/R**	**R**			**R**			**C**
8. Food for homebound clients			**C**								**A/R**	**R**	**I**

What the chart shows: **County DEM is accountable for three rows** (4, 5, and 6), and row 6 decides who does the work on rows 1, 3, and 5 in the unincorporated pockets. **Redwood City CERT is accountable for the two highest-ranked rows**, so a volunteer programme carries most of the life-saving work. **Rows 4, 5, and 6 need a decision written down, not money.** "County fire authority" is a placeholder until action 6 names the agency.

WHERE MONEY DOES THE MOST

FUNDING

The Philanthropist lens, reconciled against the ranked actions and verified. Placements are ranked by marginal lives saved per dollar. The envelope is bounded by what this place can spend before the peak, not by what it needs. No placement carries a measured cost per death averted; amounts are bottom-up unit costs, labelled as estimates.

FLOOR	TARGET	CEILING	LARGEST UNCERTAINTY
$15,000	$43,980	$110,460	Whether any PG&E rebates are left, and whether CID will run a storm pool

Floor: CERT rebate drive $5,000 + CERT and WPV-Ready/EPiC rosters $10,000 = $15,000. **Target:** $15,000 + CID storm battery pool $28,980 (20 stations × $1,199 + $5,000) = $43,980. **Ceiling:** $15,000 + a 40-station pool $57,960 (two 20-station tranches of $28,980) + Rebuilding Together Peninsula $37,500 (25 homes × $1,500, estimate) = $110,460. **The target fell 46% in verification**, from $81,480 to $43,980. The

* Suggested formulation: check with a health worker before sharing.

Rebuilding Together placement moved to the ceiling because no evidence was found that it does drainage or hazard-tree work. The ceiling fell from $172,960, because a $25,000 displacement grant had no named recipient and a second Rebuilding Together tranche had no confirmed capacity.

RECIPIENT	USE	AMOUNT	MECHANISM	BASIS	VERDICT
Redwood City CERT (with County OES)	Action 2: help device households claim the PG&E rebate	$5,000 (floor)	Grant, released only after PG&E confirms rebates remain	Estimate; no published unit cost. CERT has handled money at this scale (2018 budget $192k)	Partially confirmed
CID San Mateo	Action 2: storm battery pool for device users the rebate misses	$28,980 (target); $57,960 (ceiling)	Restricted grant, one county-wide pool	20 × $1,199 + $5,000; conditional on CID agreeing to run one pool with Belmont's	Partially confirmed; conditional
Redwood City CERT, and Woodside Fire's EPiC (WPV-Ready)	Action 1: rosters, contacts, and car buddies, split at the City–County line	$10,000 (floor)	Grants, $5,000 each	Estimates; no published basis	Partially confirmed
Rebuilding Together Peninsula	Action 3: drainage and hazard-tree work for older owners	$37,500 (ceiling only)	Restricted grant, if RTP confirms it does this work	25 homes × $1,500 (estimate). FY2025 expenses $3.35M; income-qualified owners only, doubtful in a 2.4%-poverty tract	Unverified

Cross-briefing note: the Belmont briefing (same county) places $86,940 and $25,000 with CID San Mateo. With this briefing's $28,980, CID would hold $140,920 across the two, about 7% of its $2.0M revenue, for a service it has never run. Fund one county-wide pool, not two parallel stocks. Food packs for action 8 come from Belmont's $40,000 Peninsula Volunteers placement, so they are not counted here.

The funder's one thing: most households here can buy their own resilience. The money that saves a life is small and narrow. It is for the few older, device-dependent people on single-access roads, and for the PG&E rebates that would otherwise expire unclaimed.

* Suggested formulation: check with a health worker before sharing.

Do not do this

CONTRAINDICATED · ADVICE THAT WOULD COST LIVES HERE

Advice that circulates for this hazard and would cost lives or waste the window here.

Run a generator indoors, in a garage, or in a hallway, or hand out generators without a CO alarm *

CONTRAINDICATED

At Sea Ranch on 11 January 2023, during storm outages, two men aged 71 were found dead with a generator in the hallway. The Sheriff said carbon monoxide was the suspected cause.

Instead: a battery station where possible; any generator outdoors, at least 20 feet from occupied buildings; a tested CO alarm with battery backup (step 5).

Discharge downspouts or diverted runoff onto a slope, or rely on sandbags to stop a slide

CONTRAINDICATED

Water poured onto a hillside is what saturates it. NRCS hillside guidance says to pipe downspouts to a paved roadside or storm drain. The sandbag point is general practice, not from that source: sandbags can steer surface water, but they do not hold a slope.

Instead: solid pipe to the street, and a geologist when cracks or seeps appear (step 3).

Use "voluntary" or "recommended" evacuation wording, or issue any order with no named shelter

CONTRAINDICATED

At Montecito in 2018, 19 of the 23 dead lived in the voluntary zone, and being in it was the main reason people gave for staying. On Woodside's Patrol Road in March 2023, a "highly recommended evacuation" went out before any evacuation site was set up. No one was harmed there; it shows the habit.

* Suggested formulation: check with a health worker before sharing.

Instead: California's Evacuation Warning and Evacuation Order terms, a plain consequence, and a shelter named before the order (action 5).

Write plans that rely on people driving out in peak wind

CONTRAINDICATED

44% of US tree-fall deaths were people in vehicles (Schmidlin 2009), and the Ladera death in March 2023 was a driver. On a hillside with single-access roads, the drive out is the risk.

Instead: move people before the wind peaks, or shelter them in place (actions 4 and 5).

Tell people "you'll get an alert"

CONTRAINDICATED

SMC Alert is opt-in. Without registering, the County says you "can't be reached for the majority of non-life threatening emergencies." Only life-threatening alerts reach unregistered phones.

Instead: "You won't get most alerts unless you sign up," and register people at the door (action 1, step 1).

Buy parametric rainfall cover for slide losses

CONTRAINDICATED

A policy that pays when a gauge crosses a rainfall threshold does not track when slopes fail. Slopes fail after cumulative saturation, often days after the peak rain, and no product, payer, or gauge has been named. This reasoning is carried from the Belmont briefing and transfers unchanged.

Instead: spend on drainage and warning (actions 5 and 7), and NFIP cover only where a creek or channel runs past the lot.

* Suggested formulation: check with a health worker before sharing.

WHAT HAS HAPPENED HERE BEFORE

THE RECORD

YEAR	EVENT	TOLL	WHAT CHANGED AFTER
3–5 Jan 1982 (neutral)	Bay Area storm; 25 deaths in landslides, including a Pacifica debris flow	33 dead regionally; 4 in San Mateo County (Pacifica detail from a secondary tally)	Not checked in this run
1982–83, 1997–98 (strong El Niño)	Long, saturating winters; in 1998 San Mateo County had the Bay Area's highest landslide cost	$55M of $158M regionally; the region's only landslide death	No local site named
19 Dec 2019	Oak falls into a Ferndale Way house in Emerald Hills after rain; it had recently been trimmed	No injuries; house red-tagged	Nothing found
31 Dec 2022 (La Niña)	Cordilleras Creek log-dam flood, about 10 ft deep in the channel	At least four Centennial homes damaged	The City "wants to" maintain the creek; no agreement found
Jan 2023 (La Niña)	Edgewood Rd (I-280 to Crestview) closed by slide debris; FEMA DR-4683 declared 14 Jan with household assistance	Road closed until Dec 2023	Debris removed; "No additional repair planned"
14–15 Mar 2023	Storm outages in Emerald Lake Hills, Farm Hill, and Emerald Hills	3,500+ PG&E customers out	Nothing found
21 Mar 2023	Tree falls on a van on Alpine Rd in Ladera, about 9 km away	1 dead (driver, 29)	Nothing found
22 Mar 2023	Woodside Patrol Rd: "highly recommended evacuation" for about 30 homes, no site set up	No harm recorded	Advisory lifted after five dry days
6 Feb 2024 (strong El Niño)	Storm outages in Emerald Hills	454 customers out; some Midpeninsula homes about 3 days	Nothing found

* Suggested formulation: check with a health worker before sharing.

YEAR	EVENT	TOLL	WHAT CHANGED AFTER
24 Dec 2025	Christmas Eve outages	1,200+ Redwood City and Emerald Lake Hills residents out about a day	Nothing found

The Historian's one thing: atmospheric rivers hurt this place in every ENSO phase. El Niño's signature is the long, saturating winter, when slope failures pile up into February and March. Plan on the run of storms, not the single forecast.

WHERE THE EXPERTS DISAGREE — UNRESOLVED

- **El Niño is not the local driver** versus **El Niño brings the long, saturating winter.** The Historian found declarations in every ENSO phase: only four of 15 since 1981 came in El Niño winters, and the deadliest storm came in a neutral one. The 1982–83 and 1997–98 winters show the long-wet-winter signal. Both hold. The plan is built on the run of storms, triggered by the soil and the forecast, not by the label.

WHAT NOBODY COVERED — BLIND SPOT

No record of any slide or road closure on Canyon Road or Farm Hill itself was found. Nobody maps which single-access driveways serve someone on oxygen. Until someone does, actions 1 and 2 are aimed at a count, not at doors. The door-knock in action 1 is how that map gets built.

HOW TO READ THIS — METHOD

The panel is author-built. Eight simulated expert lenses were used to generate these recommendations. Where they agree, that is a strong hypothesis, not a consensus of the field. Every recommendation was then independently checked for evidence, transferability to this location, and feasibility before the peak; the verdict tags carry the result.

Mortality figures are resolved at Climate Impact Lab impact-region scale, roughly 300,000 people. They are not neighbourhood figures. The El Niño projection covers *heat only*; falling trees, outages, slides, and creek flooding are additional to it and are carried here by the historical record rather than by a published number.

* Suggested formulation: check with a health worker before sharing.

- **Scope.** The request gave a street address in the City of Redwood City. A street address is too fine, so the briefing covers Redwood City's Canyon neighbourhood and the Farm Hill hillside, with the unincorporated County pockets inside Canyon and Emerald Lake Hills to the west. Tract figures describe tract 6098; emPOWER figures describe ZIP 94062 (75.8 square miles, Medicare only). The CPUC fire-threat tier comes from a point query of the CPUC layer at the address. Climate Impact Lab impact region, as this report records it: San Mateo County, CA, inferred rather than resolved on the Impact Map. Mortality figures exist only at that scale (impact regions average roughly 300,000 people); sub-city detail is used for hazard exposure and action design, not for death counts.
- **Exposure.** Climate Impact Lab, *451,000 at Risk* (September 2026, pre-peer-review): ECMWF SEAS5 seasonal forecasts (51-member ensemble) applied to temperature–mortality relationships across 24,378 regions, against a 1996–2025 baseline, for June 2026 – February 2027. It covers heat only. It is never added to the Impact Map's long-run projections or quoted from the map's GDP-denominated mortality. Heat is not this place's winter hazard, so the historical record carries the hazard instead.
- **Expert panel.** Eight simulated expert lenses (emergency manager, public health scientist, frontline responder, skeptic, economist, historian, philanthropist, and food supply chain analyst) each researched this place independently and in parallel, from web sources, to a fixed brief. The Belmont briefing (same county) was used as an evidence pack, and its claims were re-checked rather than inherited.
- **Reconciliation.** Duplicate actions were merged and backing lenses recorded. Convergence was tested for shared bias (whether the evidence behind an agreed action came from a setting like this one) before it counted in an action's favour. Contradictions were kept, not resolved by dropping a side. Actions were ranked by expected lives saved per unit of effort before the peak, each was given a named payer and trigger or marked unfunded, and the funding placements were reconciled against the ranked list.
- **Verification.** Three separate verifier agents checked the 8 actions, the 6 contraindicated items, the 5 resident steps, the historical record, and the funding placements, on evidence, transferability, and feasibility. Financial actions were also checked for availability in this jurisdiction, a named payer, a trigger threshold, and whether they already exist. Verifiers queried the CPUC fire-threat layer, HHS emPOWER, the Census geocoder and ACS, and OpenFEMA declarations directly, and read IRS filings on ProPublica. The envelope was re-summed.
- **Who does what.** The RACI chart is built from the owners named in the verified actions, with one accountable party per action. It is a proposal: no organisation named in it has been contacted or has agreed to a role.
- **Verdicts.** *Confirmed*: evidence, transferability, and feasibility all hold. *Partially confirmed*: holds after the corrections shown. *Unverified*: could not be checked to a primary source. *Contraindicated*: would do harm or waste the window here.
- **Adaptation Inventory.** Not consulted for this location. No action here is marked as having or lacking a precedent there.

* Suggested formulation: check with a health worker before sharing.

- **Reader.** No role was given, so the For you section is written for a resident. The lenses were briefed before every briefing was required to carry a For you section, so the resident steps were built in reconciliation from what the panel found, and verified separately on the same three tests (1 confirmed, 4 corrected). The rest of the report is written for the city, the county, and their partners.
- **Health wording.** Passages where a wording error could hurt someone (medical equipment in an outage, and generators and carbon monoxide) are marked "* Suggested formulation: check with a health worker before sharing." They were checked against published sources, not by a health professional. They are suggestions, not medical advice, and anyone passing them on should confirm them with a doctor, pharmacist, nurse, or community health worker first. The marker was added on 2 November 2026.
- **Rules.** No cost is derived by dividing a published total, and no local figure is invented; where data is thin, the report says so. Every source is listed below with its verification status.
- **Limits.** The panel and verifiers are AI research agents working from web-available sources, not human experts, and no one in this place was interviewed. The impact region was not resolved on the Impact Map. Almanac and San Mateo Daily Journal pages could not be read, so search snippets were used for the Patrol Road advisory end and the 2024 Cordilleras Creek status. OpenFEMA NFIP endpoints returned 404, so no local flood-policy count is given. The County fire-jurisdiction map was not checked, and the Genasys zone for the address was not found. ENSO phases are partly from memory of NOAA's ONI table.

References · Census Reporter, tract 6098 (ACS 2020–24) (verified) · HHS emPOWER, ZIP 94062 (queried) · CPUC High Fire-Threat District layer (queried) · PG&E Generator and Battery Rebate (verified) · PG&E Portable Battery Program (verified, closed for 2026) · PG&E Medical Baseline (corrected) · PG&E DDAR fact sheet (verified) · CID San Mateo (verified) · CPUC, CARE and FERA (verified) · SMC Alert (verified) · San Mateo County Genasys zones (verified) · WPV-Ready (verified) · Redwood City CERT (verified) · Walking Redwood City, Canyon (blog, not confirmed on a County map) · Lurie et al. 2015 (verified) · Schmidlin 2009 (verified, abstract) · KTVU, Emerald Hills oak, 2019 (verified) · ABC7, Emerald Hills oak (verified) · RWC Pulse, Ladera death and Patrol Rd (verified) · Almanac, Patrol Rd advisory ends (snippet only) · RWC Pulse, March 2023 outages (lens) · Montecito warning survey (secondary) · UCSB, Goto et al. 2021 (verified) · USGS fact sheet FS-0112-95 (verified) · RWC Pulse, Cordilleras Creek flood (corrected) · OneShoreline 2024 maintenance filing (verified) · County Public Works, Edgewood Rd (corrected) · KTVU, Sea Ranch deaths (verified) · CBS, Sea Ranch deaths (verified) · NRCS hillside fact sheet, 2017 (verified) · CRS, NFIP authorization (verified) · FEMA, mudflow and the NFIP (verified) · CDI fact sheet, Feb 2025 (verified) · OpenFEMA, DR-4683 (queried) · OpenFEMA, San Mateo County declarations (lens) · USGS PP 1434, January 1982 storm (lens) · January 1982 tally, Pacifica (secondary) · USGS MF-2325, 1997–98 landslides (lens) · San Mateo County, CalFresh replacement (verified) · CalFresh mass replacement, Jan 2023 (verified) · Almanac, Feb 2024 outages (lens) · Almanac, Christmas 2025 outages (lens) · Rebuilding Together Peninsula, IRS filings (verified; scope unverified) · NOAA ONI table (partly from memory)
Exposure data · Climate Impact Lab, 451,000 at Risk · Impact Map · Adaptation Inventory
Generated 26 October 2026 for Canyon, Redwood City. El Niño Ready is volunteer-run and is not affiliated with Climate Week NYC, The Climate Group, or Columbia University.

* Suggested formulation: check with a health worker before sharing.